PRODUCT CASE STUDY · PRODUCT INTERN ASSIGNMENT

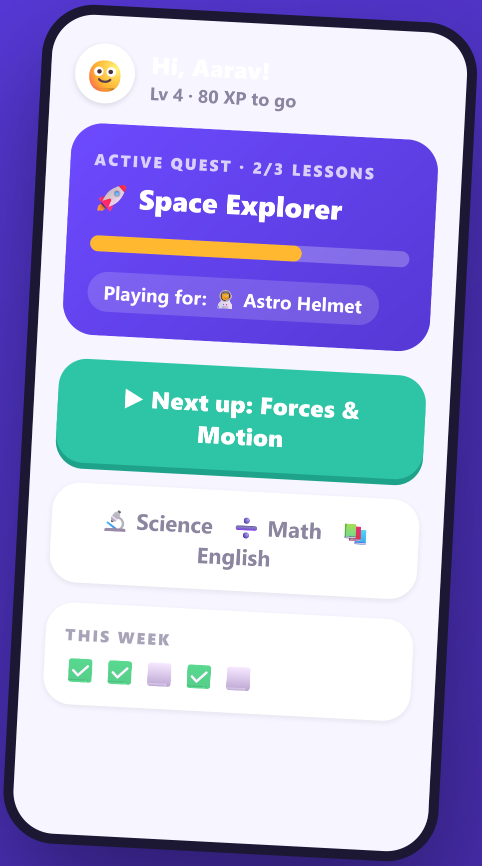
Learnify: Reducing the Lesson-3 Drop-off

Why students quit after three lessons - and one feature to change it.

🗺️ Feature: Quests

🤖 AI: personalized suggestions

🧪 Validated with an A/B test



THE PROBLEM

70% of students stop after just three lessons

The drop happens before a habit can form - and it happens even though badges, levels and leaderboards already exist.

Students say rewards "aren't motivating enough." The real question: why do they stop working after lesson 3?

BEING HONEST ABOUT THE DATA

What we know vs. what we believe

FACTS

- 70% drop off after completing three lessons
- Signups are enthusiastic - acquisition isn't the problem
- Students say rewards aren't motivating enough

ASSUMPTIONS - plausible, unproven

- Early badges come fast; then the next reward feels far away
- Leaderboards only motivate the top few
- After onboarding, there's no clear "next thing to work toward"
- No strong reason to come back *tomorrow*

Key read: "rewards aren't motivating" is a *symptom students can articulate*. The cause may be the goal structure - not the size of the rewards.

PRODUCT HYPOTHESIS

The rewards point backwards. Nothing points forwards.

Today: badges & levels

Certify what you already did. After lesson 3, the next one is a long grind away.

Missing: a near goal

Something the student *chose*, only a few lessons away, that makes lesson 4 worth doing today.

Students drop off after lesson 3 because the app stops giving them a meaningful short-term goal. Give every student a small, self-chosen goal with a reward they picked - and more of them will start lesson 4 and come back tomorrow.

THE FEATURE - ONE, NOT FIVE

Quests: short goals with a reward you picked

Pick a quest

3 suggestions, each with a "why this one for you"

→

Pick your reward

Known upfront - anticipation, not gambling

→

3-5 lessons

Finishable in days; progress fills after every lesson

→

Celebrate → next pick

There is never a moment without a next goal

Why not the obvious alternatives?

Mystery boxes = slot-machine mechanics for 12-year-olds, and no goal. Friend challenges = cold start + the leaderboard already shows competition isn't retaining the middle. Streaks = guilt when you miss a day.

Education guardrails, by design

Rewards are cosmetic only - learning content is never locked. Quests don't expire. No nag notifications, no streak pressure, solo by default.

AI PRODUCT THINKING

AI picks. Humans write.

What AI does	Ranks which 3 quests to suggest per student, and at what difficulty - using lesson history, accuracy by topic, pace, and past quest/reward picks. Surfaces a "why" line with every suggestion.
What rules do	Progress tracking, reward economy, celebrations, caps. None of this needs AI - so none of it uses AI.
Why AI beats rules here	"Always suggest the weakest subject" turns quests into homework; "always the strongest" never stretches anyone. The right next goal differs per student and shifts weekly.
Safety	No AI-generated free text ever reaches a child - quests and encouragement lines are teacher-written; AI only selects. Bad suggestion = low cost (student picks from 3; defaults exist). Only in-app learning signals, no personal data.

Honest note: Quests works with simple heuristics in v1. AI improves the feature - it isn't a crutch holding it up.

USER JOURNEY

Aarav, grade 7 - never wondering "why bother?"

Opens app

Quest card front and center: 2/3 lessons, the helmet he picked

One tap

"Next up: Forces & Motion" - no browsing, no decisions

Finishes lesson

+30 XP, quest bar animates 2/3 → 3/3

Quest complete

Astro Helmet unlocked - the reward he chose, earned

Picks the next

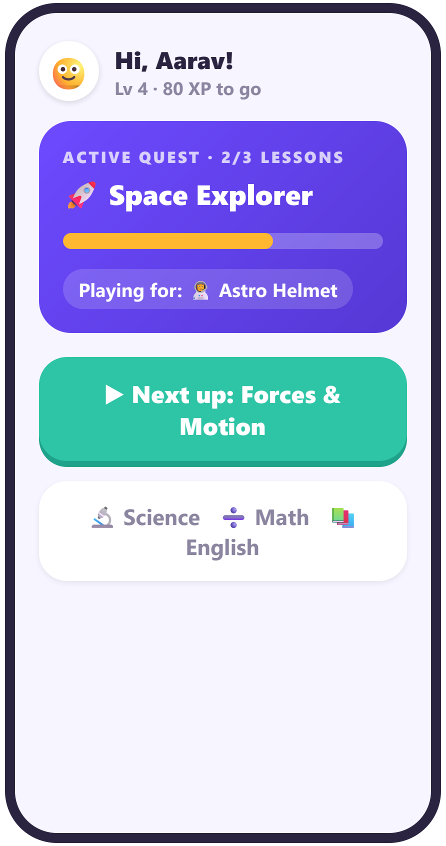
3 suggestions with "why" lines; picks Number Ninja

Comes back

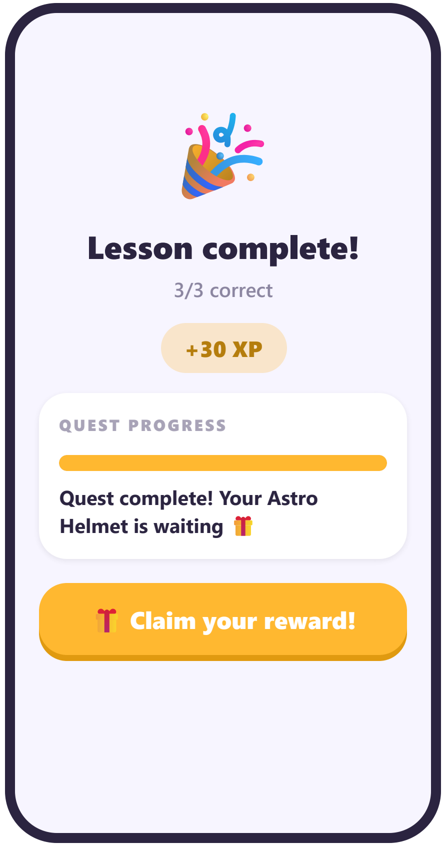
Tomorrow has a concrete goal waiting: 0/3, Robo Sidekick at stake

The loop's job: at any moment, Aarav can answer "*what am I working toward, and why do I care?*" - and the answer is days away, not weeks.

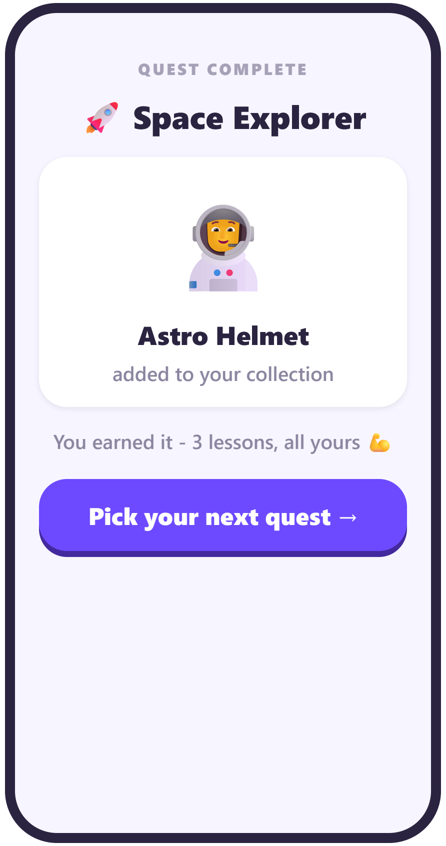
Four moments that carry the loop



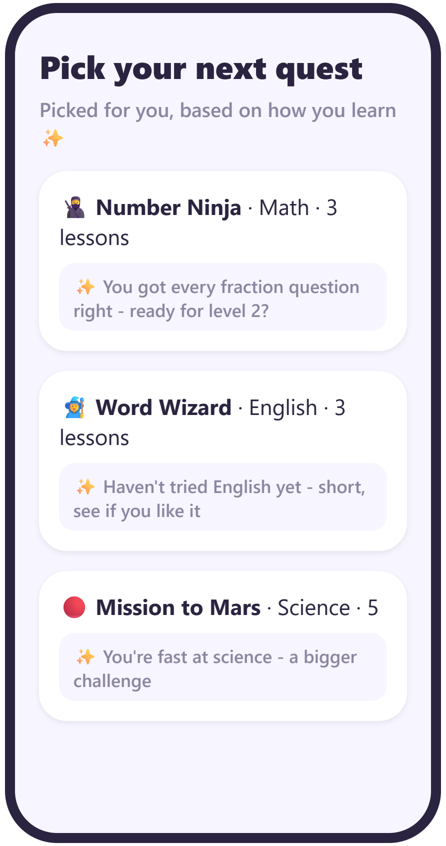
Home - the goal owns the screen



Progress you watch happen



Chosen reward, earned



AI suggests, student chooses

MEASUREMENT

Three metrics, three guardrails

PRIMARY

Lesson-4 conversion

% of new students completing a 4th lesson within 14 days. This *is* the cliff - if Quests works, it moves first.

HABIT

D7 retention

Short quests should create a reason to return tomorrow. D7 shows a habit forming, not just a longer first session.

DEPTH

Lessons per student, first 14 days

Learning depth, not just survival. Flat here + up elsewhere = students poking at UI, not learning.

Learning-quality guardrails (must not degrade): quiz accuracy per lesson · mid-lesson abandonment · time per lesson. More sessions with worse accuracy is a **failure**, even if the retention chart looks great.

A/B TEST & RECOMMENDATION

One test, clear decision rules

SETUP

New signups only, 50/50 at account creation, stratified by grade. Control: current app. Experiment: + Quests after the first completed lesson.

SIZING & DURATION (assumptions labeled)

Baseline lesson-4 conversion ~30%; detect **+5pp** at 80% power → ~1,400/arm. At ~500 signups/week: ~6 weeks enrollment + 14-day windows ≈ **8 weeks**.

SHIP / ITERATE

Ship if +5pp significant with flat guardrails. **Iterate** if positive but short - find where the loop leaks.

RETHINK / FIX FIRST

Rethink if flat despite >60% adoption - the hypothesis was wrong, and that's worth knowing. **Fix discovery first** if adoption <40% - we tested a feature nobody saw.

Recommendation: build Quests and test it with new signups. It attacks the most plausible root cause - no meaningful short-term goal - with one contained feature, uses AI only where personalization genuinely needs it, and the test tells us clearly whether the hypothesis was right.